

SIPP MIDTERMS

TOPIC 5 – INTRODUCTION TO PRIVACY ISSUES

Privacy – The right to control personal information and freedom from unauthorized intrusion.

Digital Privacy – Privacy concerns in the digital age caused by data collection and surveillance.

TYPES OF PRIVACY

- **Informational Privacy** – The right to control personal information. It protects against unauthorized access, collection, use, or sharing of personal data. Common in digital systems such as social media and banking apps.

Example: Data Privacy Act and GDPR protect informational privacy.

- **Bodily Privacy** – Protects the integrity and autonomy of the human body. It prevents forced medical procedures, biometric data collection, or physical surveillance.

Example: Consent before medical tests or body searches.

- **Communicational Privacy** – Ensures the privacy of personal communication such as calls, messages, and emails. It protects against eavesdropping, interception, and unauthorized recording.

Example: Encrypted messaging applications such as Signal or WhatsApp.

- **Territorial Privacy** – Protects an individual's physical environment or personal spaces such as homes, offices, and private property. It prevents unauthorized intrusion by people, cameras, or monitoring tools.

Example: Search warrants are needed before entering private property.

PRIVACY PROTECTION AND THE LAW

GDPR – General Data Protection

Regulation. A privacy law in the **European**

- Union and one of the strictest and most comprehensive privacy laws globally.
- **Requires clear consent** before collecting personal data; gives individuals the “right to be forgotten”; organizations must report data breaches promptly.
- **Right to be Forgotten** – The right of an individual to request that personal data be deleted.

CCPA – California Consumer Privacy Act. A privacy law in **California**, United States that gives consumers more control over their personal information.

- Right to know **what personal data is collected**; right to opt out of the sale of personal data; right to access and delete personal information; focuses on transparency and consumer control.
- **Opt Out** – Choosing not to participate in or allow a particular activity, such as the sale of personal information.

HIPAA – Health Insurance Portability and Accountability Act. A **United States** law that protects medical and health-related information.

- Sets rules for storing, accessing, and sharing patient records; requires healthcare providers and insurers to safeguard health data.

Example: A hospital cannot share a patient's medical records with an unauthorized third party without proper consent.

PDPA – Personal Data Protection Act.
Regulates the collection, use, and disclosure of personal data.

- Organizations must obtain consent before collecting data; individuals have the right to access and correct their data; encourages proper data protection practices.

Region: Singapore, Malaysia, and Thailand.

KEY PRINCIPLES OF PRIVACY LAWS

- **Consent** – Individuals must actively agree before their personal data is collected, used, or shared. Consent must be freely given, specific, informed, and unambiguous.

Example: A mobile app asks permission to access a user's location before using GPS services.

- **Transparency** – Organizations must clearly inform users about what data is collected, how it will be used, who it will be shared with, and why.

Example: A website's privacy policy explains what information it collects, why it collects it, how long it stores it, and who it shares it with.

- **Data Minimization** – Organizations should only collect the minimum amount of personal data necessary to fulfill a specific purpose.

Example: An online survey asks only for age and opinion instead of unnecessary personal information.

- **Right to Access/Delete** – Individuals have the right to access personal data held by an organization and request deletion when it is no longer needed or when consent is withdrawn, unless there is a legal reason to retain it.

KEY PRIVACY ISSUES

- **Surveillance & Government Spying** – Government monitoring of individuals' online or offline activities,

often without their knowledge or consent.

- **Corporate Data Mining** – The practice where companies collect, analyze, and sometimes sell users' personal data for advertising, profit, or profiling purposes.
- **Facial Recognition** – Technology that uses a person's facial features for identification.
- **Biometrics** – Physical or behavioral characteristics used to identify a person, such as facial scans, fingerprints, or iris patterns.
- **Facial Recognition & Biometrics Issue** – The use of biometric data for identification and surveillance can raise concerns about misuse and consent.

ANONYMITY ISSUES

- **Dark Web** – A hidden part of the internet accessible only through special browsers. It allows users to remain anonymous.
- **Dark Web Issue** – Although it can protect privacy and freedom of speech, it can also be used for illegal activities such as drug trafficking, weapon sales, and human trafficking.
- **Anonymous Social Media** – Platforms or accounts that allow users to post and interact without revealing their real identity.
- **Anonymous Social Media Issues** – Anonymity can lead to cyberbullying, online harassment, fake news, and fake accounts used to deceive or manipulate others.
- **Cryptocurrency & Privacy** – Cryptocurrencies such as Bitcoin or Monero can allow users to make financial transactions without revealing their identity.

- **Cryptocurrency Privacy Issues** – Financial privacy can also create opportunities for money laundering, illegal trade, and tax evasion because of limited traceability.

PROTECTING YOUR PRIVACY

- Use strong passwords and 2FA.
- Limit social media sharing.
- Encrypt communications using secure services such as Signal or ProtonMail.
- Read privacy policies before agreeing.

TOPIC 6 – FREEDOM OF EXPRESSION **FREEDOM OF EXPRESSION**

- The **right of every person to express thoughts, opinions, beliefs, and ideas** without fear of censorship or punishment, provided that the expression does not violate the rights of others or break the law.

FREEDOM OF EXPRESSION INCLUDES

- Speaking and sharing opinions.
- Writing articles, books, or blogs.
- Expressing ideas through art, music, or performances.
- Participating in peaceful protests.
- Using social media to communicate.
- Receiving and sharing information.

WHY IS FREEDOM OF EXPRESSION IMPORTANT?

Encourages Democracy – People can express opinions about government policies and participate in elections and public discussions.

Promotes the Exchange of Ideas – People learn from different viewpoints, which can lead to innovation and better solutions.

Protects Human Rights – People can speak against discrimination, injustice, or corruption.

Encourages Creativity – Artists, musicians, writers, filmmakers, and

inventors can express creativity through different forms of media.

Supports Academic Freedom –

Students and teachers can ask questions, conduct research, and discuss ideas openly.

WHY IS FREEDOM OF EXPRESSION NOT ABSOLUTE?

- Because **everyone has rights that must be respected**. Expression may be limited when it violates the rights of others, breaks the law, or threatens public safety and order.

LIMITS TO FREEDOM OF EXPRESSION

Hate Speech – Expressions that promote **hatred, discrimination, or violence** against a person or group because of characteristics such as race, religion, nationality, gender, disability, or ethnicity.

Public Order – Maintaining peace, safety, and order within society.

Privacy – The right of individuals to keep their personal information confidential.

National Security – **Protecting the country from threats** such as terrorism, espionage, and armed conflict.

Defamation – Making false statements that damage another person's reputation.

Libel – **Written or published** defamatory statements. Examples include statements in newspapers, websites, and social media posts.

Slander – **Spoken false statements** that damage another person's reputation. Examples include conversations, speeches, meetings, and verbal rumors.

Obscenity – Materials or acts that are grossly offensive, sexually explicit, and violate accepted standards of decency.

Child Exploitation – Using, abusing, or exploiting children for sexual, commercial, or other harmful purposes.

TOPIC 7.1 – COMPUTER CRIME, INTERNET CRIME, AND SECURITY INCIDENTS

Cyberbullying

- Bullying done through digital technologies such as social media, messaging apps, online games, and mobile phones. It often involves repeated actions meant to scare, shame, anger, or hurt someone.

Examples

- Posting embarrassing photos without permission
- Sending repeated insults or threats
- Spreading rumors online
- Excluding someone from online groups
- Creating fake accounts to mock someone

Online Harassment — Harmful online behavior such as threats, insults, humiliation, or repeated unwanted messages.

Trolling — Posting comments mainly to provoke anger, confusion, or conflict.

Example: Repeatedly commenting insulting statements on someone's post to provoke a reaction.

Cyberbullying vs. Online Harassment vs. Trolling

- **Cyberbullying** — Usually repeated and targeted harm.
- **Online Harassment** — Unwanted abusive behavior online.
- **Trolling** — Provoking others for reaction or entertainment.

Important: Even if the sender says “joke lang,” the impact on the victim still matters.

Cyberbullying vs. Online Harassment vs. Trolling

- **Cyberbullying** — Usually repeated and targeted harm.
- **Online Harassment** — Unwanted abusive behavior online.
- **Trolling** — Provoking others for reaction or entertainment.

Important: Even if the sender says “joke lang,” the impact on the victim still matters.

Effects of Harmful Online Behavior

- Anxiety or fear
- Loss of confidence
- Isolation from classmates
- Poor academic performance
- Damaged reputation
- Conflict in school or community

Important: Digital actions can leave long-term digital footprints.

Misinformation — False or inaccurate information shared without intent to harm.

Disinformation — False information deliberately created or shared to mislead or manipulate.

Fake News — False or misleading content made to look like real news.

Example of Misinformation — A post says, “Classes are suspended tomorrow. Share this now!” but there is no announcement from the school or official page.

What should you do? Check official school channels first before sharing.

How False Information Spreads

- Sharing without reading the full post
- Believing emotional headlines
- Trusting screenshots without checking sources
- Sharing posts from friends without verification

- Reacting quickly because of fear or anger

Responsible Sharing Checklist

Before posting or sharing, ask:

- Is it true?
- Is the source reliable?
- Is it updated?
- Could it harm someone?
- Do I have permission to share it?
- Am I helping or just adding noise?

Digital Citizenship — Using technology responsibly, safely, respectfully, and ethically.

A Good Digital Citizen:

- Respects others online
- Protects privacy
- Thinks before posting
- Verifies information
- Reports harmful behavior
- Uses technology for positive purposes

Harmful Post — Online content that can negatively affect or cause harm to another person.

Misleading Post — Content that gives false or inaccurate information and may cause people to believe something untrue.

Abusive Post — Content involving insults, humiliation, threats, or behavior that violates or harms another person.

Responsible Post — Online content that is truthful, respectful, verified, and shared with consideration for others.

Digital Footprint — The lasting record or trace created by a person's digital actions and activities online.

Privacy Violation — Sharing someone's private information or private messages without permission.

Example: Sharing a screenshot of another student's private message with

insulting captions is abusive and harmful because it violates privacy and humiliates the person.

TOPIC 7.2 – INTELLECTUAL PROPERTY, COPYRIGHT, PLAGIARISM, AND SOFTWARE PIRACY

INTELLECTUAL PROPERTY

- Intangible creations of the mind. It includes inventions, literary and artistic works, designs, symbols, names, and images used in commerce such as logos and trademarks.

IMPORTANCE OF INTELLECTUAL PROPERTY

Protection for Creators – Grants legal rights to authors, inventors, and businesses over their creations and prevents unauthorized use, reproduction, or exploitation.

Encourages Innovation and Creativity – Provides incentives for individuals and companies to create new ideas and products.

Economic Growth and Fair Competition – Gives creators recognition and financial benefits, encourages healthy competition, and improves product quality.

Cultural and Social Development – Protects cultural heritage and promotes the dissemination of knowledge while respecting creators' rights.

TYPES OF INTELLECTUAL PROPERTY

Copyright – Legal protection given to creators of original literary, artistic, musical, and other creative works.

Covers – Books, songs, films, paintings, software, architecture, and other creative works.

Key Points – Gives exclusive rights to reproduce, distribute, and display the work.

Protects the expression of an idea, not the idea itself. Duration varies by country and is

often based on the creator's lifetime plus a set number of years.

Patent – **Exclusive rights** granted to inventors for a new and useful invention, process, or design.

Covers – New machines, industrial processes, chemical formulas, and technological innovations.

Key Points – **Gives the right to exclude others from making, using, or selling the invention without permission.** The invention must be novel, non-obvious, and useful. A patent is typically valid for 20 years, depending on jurisdiction.

TRADE SECRET

Trade Secret – Confidential business information that provides a competitive advantage.

Covers – Recipes, formulas, strategies, methods, customer lists, and similar confidential information.

Key Points – Must remain secret. Once publicly disclosed, protection is lost. No formal registration is required; protection depends on efforts to maintain secrecy such as NDAs and security measures.

Examples – Coca-Cola's formula and KFC's blend of herbs and spices.

IMPORTANT COMPARISON

Copyright – Protects original creative works.

Patent – Protects new and useful inventions, processes, or designs.

Trade Secret – Protects confidential business information.

KEY INTELLECTUAL PROPERTY ISSUES

Infringement – Unauthorized use of another person's protected intellectual property without permission.

Examples – Copying music, movies, or books without the creator's consent; producing counterfeit products such as fake branded clothing or electronics.

Enforcement – The process or effort of protecting and applying intellectual property rights.

IP Enforcement Issues – IP laws vary from country to country, making enforcement complex. Problems include lack of resources or legal mechanisms in some regions and expensive, time-consuming legal processes.

Digital Piracy – Illegal sharing, copying, downloading, or distribution of copyrighted digital content.

Examples – Illegal downloading or streaming of movies, software, or e-books; sharing copyrighted materials on social media or torrent sites.

Balancing Rights – Finding a **balance** between creators' rights and the public's access to knowledge.

Fair Use – **Limited use** of copyrighted material for purposes such as education, research, commentary, or parody.

Creative Commons (CC) – **Free legal tools** that allow creators to share their work legally under selected permissions. Instead of “all rights reserved,” Creative Commons allows creators to choose “some rights reserved.”

Works that may use Creative Commons licenses include photos, videos, music, and educational materials.

WHY CREATIVE COMMONS IS IMPORTANT FOR STUDENTS

Creative Commons helps students determine:

- Whether a work can be reused.
- Whether it can be edited or remixed.
- Whether it can be used for school projects.
- Whether credit must be given to the creator.

- Whether it can be used for commercial purposes.

Plagiarism – Using someone else's words, ideas, images, code, or work without proper credit. It can occur in essays, reports, PowerPoint presentations, programming activities, research papers, infographics, videos, and online posts.

TYPES OF PLAGIARISM

Direct Plagiarism – Copying exact words from a source without quotation marks or citation.

Example: Copying a paragraph from Wikipedia and placing it directly in a report.

Self-Plagiarism – Reusing your own previous work for a new requirement without permission.

Example: Submitting the same essay in two different subjects.

Mosaic Plagiarism – Mixing copied phrases from different sources and presenting them as original work.

Example: Changing only a few words from an online article.

Accidental Plagiarism – Forgetting to cite a source or using an incorrect citation.

Citation – Giving proper credit to the original source of information.

Students should cite sources when using:

- Exact words from another author.
- Ideas from a book, website, video, or article.
- Images, charts, or tables.
- Code or algorithms.
- Research data.
- Definitions or explanations from another source.

Academic Honesty – Responsible and truthful use of information and proper acknowledgment of sources in academic work.

HOW TO AVOID PLAGIARISM

Take notes carefully – Write down the source while researching.

Use quotation marks – Use quotation marks when copying exact words.

Paraphrase properly – Use your own words and sentence structure.

Cite the source – Give credit even when paraphrasing.

Add your own analysis – Do not only copy information; explain it using your own understanding.

Keep a reference list – List all sources used at the end of the work.

License – A **permission agreement** that explains how a person may use a work, product, or software.

Licenses may apply to software, images, music, videos, fonts, templates, e-books, and online learning materials.

- Whether users can copy the work.
- Whether users can edit or remix it.
- Whether users can share it with others.
- Whether users can use it commercially.
- Whether users must give credit.
- How many users may install or access it.

Key Idea – Buying or downloading something does not always mean that the user can do anything they want with it.

SOFTWARE LICENSING

- Software is protected by copyright, and users must follow the license agreement given by the software owner.

Freeware – Software that is free to use but may still have restrictions.

Open-Source Software – Software whose source code is available under certain conditions.

Proprietary Software – Software owned by a company or developer that usually requires payment or permission.

Trial Software – Software that is free only for a limited time.

Subscription Software – Software that requires regular payment to continue using it.

SOFTWARE PIRACY

Software Piracy – Unauthorized copying, downloading, sharing, or use of software in violation of its copyright or license. Important idea: Software is protected by copyright, so users must follow the software owner's license agreement.

TOPIC 8 – SOFTWARE DEVELOPMENT

Software Development – The systematic process of designing, coding, testing, deploying, and maintaining software applications or systems.

Software Development Life Cycle (SDLC) – A structured process used to develop software through different phases to ensure that the final product meets user requirements and performs reliably.

IMPORTANCE OF SOFTWARE DEVELOPMENT

Enhances User Satisfaction – Well-developed software provides a seamless and efficient user experience, improving customer satisfaction and loyalty.

Supports Business Success – Quality software helps organizations streamline operations, increase productivity, and maintain a competitive advantage.

Ensures Safety and Reliability – Software reliability and security are important in critical sectors such as healthcare, transportation, and finance to prevent failures that could cause harm or losses.

Drives Innovation – Software development supports the creation of new technologies and solutions.

Economic Impact – High-quality software reduces maintenance costs, minimizes downtime, and contributes to business profitability and sustainability.

STRATEGIES FOR ENGINEERING QUALITY SOFTWARE

Requirement Analysis – Gathering and analyzing user and business needs to define clear and measurable requirements. Proper analysis **helps prevent misunderstandings and costly rework.**

Design – Creating system architecture and detailed design specifications, including data models, user interfaces, and workflows. It **focuses on scalability and maintainability.**

Implementation (Coding) – Writing clean, modular, and efficient code that follows defined standards.

Testing – Conducting thorough testing to identify and fix defects before deployment. **Includes functional, performance, and security testing.**

Deployment – Delivering the software to users in a stable and reliable manner, often through automated pipelines.

Maintenance – Regularly updating and improving software to fix bugs, add features, and ensure compatibility with evolving technologies.

BEST PRACTICES

Modular and Reusable Code – Breaking a system into smaller, independent modules that can be reused across projects. This improves maintainability and reduces redundancy.

Coding Standards and Guidelines – Rules involving naming conventions, documentation standards, and design principles such as SOLID and DRY to maintain consistency and readability.

Version Control System – A system used to track changes, collaborate effectively, and manage different versions of software.

Git – An example of a version control tool.

QUALITY ASSURANCE (QA)

Quality Assurance (QA) – Ensuring software quality through continuous and comprehensive testing.

Unit Testing – Verifying that individual components or modules work correctly.

Integration Testing – Ensuring that different modules interact seamlessly.

System Testing – Validating the entire software product against its requirements.

Continuous Testing and Automation – Automating repetitive test cases to save time and detect issues early during development.

PEER REVIEWS AND CODE

INSPECTIONS

Regular Code Review – Peer review during development that helps detect defects early, improve code readability, and promote knowledge sharing.

Formal Inspection – A systematic and documented evaluation of code or design to ensure compliance with standards and uncover subtle errors.

DOCUMENTATION

Documentation – Detailed information about requirements, architecture, APIs, and user manuals.

Importance of Documentation – Helps onboard new developers, makes troubleshooting easier, and supports scalability and future improvements.

AGILE AND DEVOPS

Agile – A software development approach focused on delivering small, working software increments frequently, allowing teams to receive feedback and adapt quickly.

Iterative Development – Developing and delivering software in small increments repeatedly.

Quick Feedback Loops – Continuous interaction with stakeholders to ensure that software remains aligned with user needs.

DevOps – Practices that integrate development and operations to support fast and reliable software releases.

CI/CD Pipelines – DevOps processes involving Continuous Integration and Continuous Deployment.

Continuous Integration (CI) – Automated builds and tests that help integrate changes frequently.

Continuous Deployment (CD) – Automatically delivering software changes to users through deployment processes.

SOFTWARE PRODUCT LIABILITY

Software Product Liability – The legal responsibility of software developers, vendors, or companies for harm or damages caused by defects or failures in their software products.

Software liability may arise when software malfunctions result in financial loss, safety hazards, security breaches, or injury to users.

TYPES OF SOFTWARE LIABILITY

Negligence – Occurs when developers fail to meet reasonable industry standards for quality and safety.

Example: Inadequate testing or ignoring known vulnerabilities before releasing software.

Strict Liability – Holds producers accountable for defects even when negligence is not proven. If a software defect causes harm, the producer may be liable regardless of intent or care.

Breach of Warranty – Liability that can arise when software fails to meet an express or implied promise or warranty. The PPT identifies this as one of the legal bases of software liability.

KEY ISSUES OF SOFTWARE DEVELOPMENT

Cost and Time Overruns – Caused by poor planning or unrealistic timelines.

Quality vs. Speed – Pressure to deliver software quickly may reduce quality.

Security Concerns – Software may contain vulnerabilities and face cyber threats.

Ethical Concerns – Includes privacy issues, misuse of software, and AI ethics.

Maintenance Challenges – Difficulty keeping software updated and compatible with new technologies.

Team Collaboration Issues – Miscommunication among developers, clients, and stakeholders.

TOPIC 9 – PROFESSIONAL ETHICS IN SOFTWARE DEVELOPMENT

Professional Ethics – Moral principles and standards that guide behavior in software engineering.

It ensures that software developers act responsibly toward clients, users, colleagues, and the public.

Importance – Unethical software practices can cause harm, security breaches, financial loss, and damage to trust.

KEY PRINCIPLES FROM ACM/IEEE CODE OF ETHICS

Public Interest – Prioritize the safety, privacy, and welfare of users and the public.

Client & Employer Responsibility – Act in the client's best interests while maintaining honesty and transparency.

Product Quality – Deliver reliable, maintainable, and well-tested software.

Judgment & Professionalism – Exercise care, diligence, and sound decision-making.

Management – Promote ethical practices within teams and projects.

Profession & Colleagues – Support peers and report unethical behavior.

Self – Maintain competence, continuous learning, and integrity.

ETHICAL RESPONSIBILITIES IN SOFTWARE DEVELOPMENT

Quality – Ensure that software meets functional and non-functional requirements.

Safety – Prevent software from causing harm to users or the public.

Honesty – Report progress, limitations, and defects truthfully.

Client Responsibility – Understand client needs and provide honest guidance.

User Welfare – Protect privacy, accessibility, and usability.

Transparency & Accountability – Document processes and decisions and take responsibility for mistakes.

Quality – Ensuring that software meets its functional and non-functional requirements.

Functional Requirements – What the software should do.

Non-functional Requirements – Qualities or constraints of the software, such as performance, security, and usability.

Safety – Preventing software from causing harm to users or the public. Software developers must consider possible risks and consequences of software failures.

Honesty – Truthfully reporting software progress, limitations, and defects. Developers should not hide known problems from clients, users, or stakeholders.

Client Responsibility – Understanding client needs and providing honest guidance. Developers should act in the client's best interests while maintaining transparency.

User Welfare – Protecting the privacy, accessibility, and usability of software users.

Privacy – Protecting users' personal information.

Accessibility – Ensuring software can be used by people with different abilities.

Usability – Ensuring software is easy and effective for users to operate

Transparency – Clearly documenting processes and decisions and being open about relevant information.

Accountability – Being responsible for decisions, actions, mistakes, and their consequences.

LINKING SDLC TO ETHICS

Requirement Analysis – Ensure client needs are realistic and safe.

Design – Consider security, accessibility, and fairness.

Coding – Follow coding standards and avoid shortcuts that introduce defects.

Testing – Conduct thorough Quality Assurance to prevent harm.

Deployment – Ensure stability and reliability.

Maintenance – Responsibly address bugs, updates, and security vulnerabilities.

ETHICS IN EACH SDLC PHASE

Requirement Analysis = Ethical requirements must be realistic and safe.

Design = Security, accessibility, and fairness must be considered.

Coding = Developers should follow standards and avoid shortcuts that cause defects.

Testing = Thorough testing should be performed to prevent harm.

Deployment = Software should be stable and reliable before release.

Maintenance = Bugs, updates, and security vulnerabilities should be handled responsibly.